

Term Project

ISE 503 – Health Analytics || Spring 2026

Overview

The project will provide the opportunity to identify a realistic health challenge and to apply analytics methods to address the problem. You will work in self-formed teams of 2-3. Each team will choose a health application area and identify the decision-making problem, **gather the relevant data**, use analytics to conduct the analysis, and interpret and communicate findings and recommendations. Findings will be communicated in the form of a report and an interview-based presentation to a fictitious health sector client.

While the choice of health application area is flexible, each team should be able to clearly articulate how the application and decision-making problem relate to the health sector. The project will reinforce both technical skills (e.g., programming, model development, model interpretation, data visualization and reporting) and “soft” skills that are very important for working in the health industry (e.g., communication, teamwork, collaboration, project management).

Timeline and deliverables:

- **Week 8 (3/4):** Each team needs to be registered with their team members and several ideas for a project. (5% of the project grade)
 - **Submit on Gradescope**
- **Week 11 (4/1):** Each team submits an interim report (1-2 pages, excluding appendices) (20% of project grade).
 - The report should describe the health application area, the decision-making problem, the data, the analytics approach, initial results, the next steps, and the expected impact.
 - **Submit on Gradescope**
- **Week 15 (4/27):** Each team submits the following deliverables (all in pdf):
 - An **executive summary** including recommendations written for a senior decision-makers (around 300 words) (15% of project grade).
 - **Submit on Gradescope**
- **Week 15 (4/29):** We will hold interviews to showcase your work (30% of project grade). Each team will be interviewed by another team. The presenting team will first be given an opportunity to present its work (7 minutes, 7 slides). The interviewing team will play the role of a key decision-maker, in charge of taking and implementing the recommendations.
- **Week 16 / Finals Week (5/6):** A **report** (around 4 pages, excluding appendices)
 - Report includes a description of the health application area, the problem, your analysis, your results, and implications for health or health industry (30% of project grade).

Where to find data sources for your project

Sources for finding **health datasets** for project (see [project dataset ideas document](#))

- **U.S. Government Health Surveys & Registries:** large, free, individual-level data on chronic disease, behaviors, and healthcare utilization: [NHANES](#), [BRFSS](#), [MEPS](#), [SEER](#), [CMS](#), [CDC WONDER](#)
- **International / Global Health:** country-level aggregated data on disease burden, life expectancy, and health systems: [WHO GHO](#), [Global Burden of Disease \(IHME\)](#), [DHS Program](#)
- **Clinical / EHR Data:** individual patient-level ICU and hospital records (free with added time for access approval / DUA): [MIMIC-IV](#), [eICU](#), [All of Us](#)
- **Educational / ML Benchmark Datasets:** smaller, clean datasets (only use those with verifiable source data): [UCI ML Repository](#), [Kaggle Health](#), [BioLINCC Teaching](#), [Medical data for machine learning database on GitHub](#)
- **Published Papers with Open Data:** datasets tied to specific studies, great for extending published analyses: e.g., Framingham Heart Study, Cleveland Heart Disease, Wisconsin Breast Cancer

Team Registration

This first assignment for the final project has 2 requirements: (1) Register your team members; (2) List project ideas including the relevant data sources.

1. Names and email addresses of members of your team
2. Your project topic. For each, please provide:
 - i. the title and 1-sentence description of the data set and
 - ii. a 1-paragraph description of your proposed project topic using this data set and why it is relevant.

Brief Interim Report

Please provide a brief report containing:

- **Main text** (1-2 pages): An introduction to the health application area, the decision making problem you are focusing on addressing, your data source and data set (including whether this is a verified source or an ‘educational’ dataset from e.g. Kaggle), the analytics approaches you are planning to take, your initial results, your next steps, and the expected impact of the health problem you can help to solve with analytics.
 - You do not need to compare to results from published papers for this interim report, although that will be necessary for the discussion section of your final report.
- **Appendices:**
 - Appendix A: Your figures and tables
 - Appendix B: Your code

Final: Presentation

Each team will have **7 minutes and 7 slides** for their final presentation. Appendix slides that support the presentation or questions that may arise can be included.

The presentation should cover:

- Health problem and motivation for addressing it
- Data and methods chosen to analyze and address this problem
- Key findings
- Recommendations for a relevant health industry decision-maker

Final: Report

The report should include a description of the health application area, the problem, your analysis, your results, and implications for practice (clinical, health industry).

Example report outline:

- **Introduction:** Introduction to the problem and health context
 - Health problem
 - Motivation and Purpose
- **Methods**
 - Data source – describe the source of the data and whether it is a verifiable source; contextualize the data within the larger problem context
 - Data preprocessing
 - EDA
 - Predictive modeling
- **Results**
 - Model results and their interpretation
 - You can include parameter values chosen
- **Conclusions:** Discussion, implications, recommendations
 - Interpret and summarize the implications of your findings
 - Compare your results to those from published papers from verifiable sources
 - Discuss the impact of your model
 - Summarize the implications for practice for your decision-maker(s) / stakeholder(s) of interest (clinical, health industry)
- **Appendices:**
 - Appendix A: Your figures and tables
 - Any display item (table, figure) included in Appendix A should be referred to specifically (e.g., Figure 1 or Table 2 rather than Appendix A) in the main text
 - Do not include tables in Appendix A that are just copy-pasted output from coding packages. All tables should be recreated to display only the relevant information correctly structured.
 - Appendix B: Your code